The drift of the Collatz cascade: balance identities, the directional low-pass, and the attenuation constant

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Abstract

The full-density program for the $3x+1$ problem — pushing the Krasikov–Lagarias exponent $\gamma_k \to 1$ — was reduced in the companion paper (the tempering law) to a single drift-direction statement: the heterogeneity profile of the edge certificate obeys a two-coefficient recurrence $\mathrm{CV}_P = a\,\mathrm{CV}_{P-1} + c\,\mathrm{CV}_{P+1}$, and $\gamma \to 1$ follows if $\theta = a/c$ stays uniformly below 1. This paper isolates the mechanism of that drift and proves every structural ingredient except one. Proved: (i) mass conservation at the edge, $3W_0 + W_2 + W_8 = 3$ exactly because $2^{\log_2 3} = 3$, forcing subcriticality $a + c \leq 1$; (ii) a balance identity — the zeroth-order scale flows of offset-magnitude are exactly equal, up-flow $= W_0 \cdot \frac{3}{4} = \frac{3}{16} = \bar{w} \cdot \frac{1}{4} =$ down-flow, so the drift is entirely a second-order effect; (iii) a one-line directional low-pass: $\min(x + c\mathbf{1}) = \min(x) + c$, so the K–L minimum passes triple-constant modes with slope exactly 1 and is 1-Lipschitz on intra-triple variation, and only the down-channel passes through the minimum. Measured: the attenuation constant κ of the down-channel decomposes exactly as $\kappa^2 = \mathrm{Var}(\bar{D})/V + 2\mathrm{cov}/V + \mathrm{Var}(R)/V$ with all attenuation in the covariance term; the correction is antisymmetric (± 0.026) and not switch-exclusive, identifying the mechanism as selection-weighted mean reversion — the rigidity of binding constraints in the K–L linear program; and $\kappa < 0.95$ at every scale and depth, with the deep-scale value flat in k: $\kappa_\infty = 0.839 \pm 0.002$, uniformly away from 1. Three negative results are reported honestly: a Gaussian reduction of κ fails, one-step coefficient reconstruction inverts the drift, and per-application mode transfer is scale-scattering — the recurrence is emergent at the fixed point, so the remaining proof must be variational. We state the one remaining lemma precisely, with an LP-duality route, and collect the program's constants: the quantitative skeleton is governed by the single constant $\ln(4/3)$. All claims are labeled proved, verified, or measured.

1 Setup: the edge system, the profile, and the reduction

Let $T(n) = n/2$ (n even), $T(n) = (3n+1)/2$ (n odd), and let $\pi_1(x)$ count the $n \leq x$ whose orbit reaches 1. The strongest rigorous density bounds $\pi_1(x) \geq x^\gamma$ come from the difference-inequality method of Krasikov and Lagarias [1]. In the companion paper [5] we verified certificates to 3-adic depth $k = 20$ ($\gamma = 0.9146$), identified the certificate eigenvectors as tempered roulette measures, and reduced the full-density statement $\gamma \to 1$ to a single open inequality about the direction of a drift. This paper is about that drift.

Notation

$\beta := \log_2 3$. CV denotes coefficient of variation (standard deviation over mean). $[3^j] := \{m \bmod 3^j : m \equiv 2 \pmod 3\}$. Every assertion below carries one of three labels. *Proved*: complete mathematical proof in the text. *Verified*: finite exact or floating-point computation, reproducible from stated parameters. *Measured*: statistical fit or extrapolation from computed data. Section 9 collects the ledger.

1.1 The Krasikov–Lagarias system at the edge

Fix $k \geq 2$ and $\lambda \in (1, 2]$. The nontrivial-tree K–L system $L_k(\lambda)$ asks for $c : [3^k] \to \mathbb{R}_{>0}$ with, for every m,

$$m \equiv 2 \pmod 9 : \qquad c(m) \leq \lambda^{-2} c(4m) + \lambda^{\beta-2}\, \bar{c}\big(\tfrac{4m-2}{3}\big)\,, \tag{1}$$

$$m \equiv 5 \pmod 9 : \qquad c(m) \leq \lambda^{-2} c(4m), \tag{2}$$

$$m \equiv 8 \pmod 9 : \qquad c(m) \leq \lambda^{-2} c(4m) + \lambda^{\beta-1}\, \bar{c}\big(\tfrac{2m-1}{3}\big)\,, \tag{3}$$

where $4m$ is reduced mod 3^k, branch targets mod 3^{k-1}, and

$$\bar{c}(t) := \min\{c(t),\ c(t + 3^{k-1}),\ c(t + 2 \cdot 3^{k-1})\} \tag{4}$$

is the minimum over the three refinements (the *triple*) of t. Feasibility implies $\pi_1(x) \geq x^\gamma$, $\gamma = \log_2 \lambda$ [1]. We call $m \mapsto 4m$ the *transport* (or *up*-) channel and the two branch maps the *down*-channel; both class maps are bijections. The *edge* is $(\lambda, \gamma) = (2, 1)$: full density. At the edge, set

$$W_0 = \lambda^{-2} = \tfrac{1}{4}, \qquad W_2 = \lambda^{\beta-2} = \tfrac{3}{4}, \qquad W_8 = \lambda^{\beta-1} = \tfrac{3}{2}. \tag{5}$$

1.2 The heterogeneity profile and its recurrence

For each trit scale P of the class label ($P = 1$ the fine end, $P = k-2$ the top), let $\mathrm{CV}_P(k)$ be the mean CV over sibling triples of classes differing only in trit P of the Perron eigenvector of $L_k(\lambda_k)$. The exact pointwise transport identity for eigenvector differences (verified at $k = 13$: correlation 1.000000, residual 2.8×10^{-4} over 15,115 samples [5]) rests on the *offset algebra*: a pure trit offset $\delta = d \cdot 3^P$ satisfies, exactly,

$$4\delta = \delta + 3\delta \quad \text{(transport: scales } P, P{+}1), \qquad \tfrac{4\delta}{3} = \tfrac{\delta}{3} + \delta \quad \text{(branch: scales } P{-}1, P). \tag{6}$$

Averaging over classes yields the two-coefficient closure (measured, fit error $\leq 1.2\%$ at $k = 13, 15, 17, 19$):

$$\mathrm{CV}_P \;=\; a\, \mathrm{CV}_{P-1} + c\, \mathrm{CV}_{P+1}. \tag{7}$$

1.3 The reduction

On the conservative line $a + c = 1$ with $c > a$ the decaying solution of (7) is $\mathrm{CV}_P \sim \theta^P$ with $\theta = a/c < 1$; heterogeneity injected at the fine end then dies geometrically before reaching the top trit, forcing $q_k \to 1$ (the min-loss parameter) and, by the analytic edge rate $d\gamma/dq = 1/\ln(4/3)$ (Theorem 19 of the program; proved, see Appendix A), $\gamma_k \to 1$. The measured θ-series is 0.8248, 0.8360, 0.8438, 0.8480, 0.8488 at $k = 13, 15, 17, 19, 20$, with increments shrinking by ≈ 0.6 per depth: $\theta_\infty \approx 0.85 < 1$ [5]. The open question is therefore:

prove $\theta = a/c < 1$ uniformly in k.

This paper establishes: the cascade cannot grow ($a+c \le 1$, proved, Section 2); the zeroth-order drift is exactly zero (proved, Section 3); the entire drift is produced by a one-way low-pass filter (structure proved, Section 4); the filter's attenuation constant is measured, mechanistically identified (Section 5), and uniform in depth (Section 6). What survives as open is a single variational attenuation bound (Section 8).

2 Proved: mass conservation forces subcriticality

Lemma 1 (Mass conservation at the edge; proved)**.** *With the edge weights* (5)*, the row masses of the three constraint types* (1)–(3) *are*

$$W_0 + W_2 = 1 \quad (m \equiv 2 \bmod 9), \qquad W_0 = \tfrac{1}{4} \quad (m \equiv 5), \qquad W_0 + W_8 = \tfrac{7}{4} \quad (m \equiv 8),$$

and their average is exactly 1*:*

$$3W_0 + W_2 + W_8 \;=\; \tfrac{3}{4} + 3 \cdot 2^{\beta-2} \;=\; \tfrac{3}{4} + \tfrac{9}{4} \;=\; 3, \qquad \textit{precisely because } 2^\beta = 3. \tag{8}$$

Proof. Direct evaluation; the final step is $3 \cdot 2^{\beta-2} = 3 \cdot 2^\beta/4 = 9/4$ iff $2^\beta = 3$. □

Corollary 2 (Subcriticality; proved within the closure (7))**.** *Every unit of difference-field mass at trit scale P is redistributed by one application of the edge-linearized operator to scales $\{P-1, P, P+1\}$ with total weight equal to the row mass of its class (offset algebra* (6) *with the weights* (5)*); the minimum* (4) *is* 1*-Lipschitz and contributes no expansion. Averaging over classes and applying the triangle inequality bounds the closure coefficients of* (7)*, after absorbing the self-term, by*

$$a + c \;\le\; 1.$$

Measured at $k = 20$: $a + c = 0.9955$, the strictness being the measured incoherence factor ≈ 0.90 between the two transport channels (the triangle inequality is an equality only for perfectly coherent recombination). Lemma 1 is the reason the drift question is well-posed at all: the identity $2^\beta = 3$ — the defining equation of the $3x+1$ problem — pins the average row mass to exactly 1. The heterogeneity cascade sits exactly *on* the conservative line; it can never grow exponentially, and the only remaining freedom is the direction of drift along that line.

3 Proved: the balance identity — zeroth-order drift is zero

One might hope the drift $c > a$ is visible already at the level of average magnitude flow between scales. It is not: the flows balance exactly, and this balance is itself a consequence of $2^\beta = 3$.

Proposition 3 (Balance identity; proved)**.** *At the edge, weight the two leakage channels of the offset algebra* (6) *by their row weights and their magnitude fractions. The up-flow of offset magnitude (scale $P \to P+1$, transport channel) and the down-flow (scale $P \to P-1$, branch channel) are exactly equal:*

$$\textit{up-flow} \;=\; W_0 \cdot \tfrac{3}{4} \;=\; \tfrac{3}{16} \;=\; \bar{w} \cdot \tfrac{1}{4} \;=\; \textit{down-flow}, \qquad \bar{w} := \tfrac{W_2+W_8}{3} = \tfrac{3}{4}.$$

The zeroth-order scale drift of difference mass is exactly zero.

Proof. Transport: $4\delta = \delta + 3\delta$ splits a magnitude-$|\delta|$ offset into components of magnitude $|\delta|$ (scale P) and $3|\delta|$ (scale $P+1$); the up-going fraction is $3/4$ of the transported magnitude $4|\delta|$, and the transport weight is $W_0 = 1/4$; hence up-flow $= \frac{1}{4} \cdot \frac{3}{4} = \frac{3}{16}$. Branch: $4\delta/3 = \delta/3 + \delta$ splits magnitude $\frac{4}{3}|\delta|$ into $\frac{1}{3}|\delta|$ (scale $P-1$) and $|\delta|$ (scale P); the down-going fraction is $1/4$. The mean branch weight per class is $\bar{w} = (W_2 + W_8)/3 = (\frac{3}{4} + \frac{3}{2})/3 = \frac{3}{4}$ by mass conservation (Lemma 1: only types 2 and 8 carry a branch term, each once among three classes); hence down-flow $= \frac{3}{4} \cdot \frac{1}{4} = \frac{3}{16}$. □

This is the third exact balance produced by the single identity $2^\beta = 3$: it holds at the level of row masses (Lemma 1), at the level of the min-loss identity [5], and now at the level of offset-magnitude flow. The consequence is sharp: *the drift $c > a$ cannot be a first-moment effect.* Whatever pushes heterogeneity preferentially downward must act through the only structural asymmetry left in the system — and there is exactly one.

4 Proved: the directional low-pass

The one asymmetry between the two channels of (1)–(3) is the minimum (4): the down-channel (branch) reads the eigenvector only through $\bar{c}$, the minimum over a triple, while the up-channel (transport) reads $c(4m)$ directly. The following one-line fact turns that asymmetry into a frequency-selective filter.

Proposition 4 (Directional low-pass; proved)**.** *For $x \in \mathbb{R}^3$ and $c \in \mathbb{R}$,*

$$\min(x + c\mathbf{1}) \;=\; \min(x) + c, \tag{9}$$

and for $x, y \in \mathbb{R}^3$, $|\min(x) - \min(y)| \le \max_i |x_i - y_i|$. Hence, decomposing any perturbation of a triple into its triple-constant (coarse) component and its intra-triple-varying (fine) component: the minimum transmits coarse modes with slope exactly 1 and is 1-Lipschitz — weakly contractive — on fine modes. Since only the down-channel passes through the minimum, the K–L edge operator transmits all modes freely upward, and transmits coarse modes freely but attenuates precisely the intra-triple-varying modes downward.

Proof. (9): every component shifts by c, so the least does. The Lipschitz bound: $x_i \le y_i + \max_j |x_j - y_j|$ for all i; take minima over i and symmetrize. □

This is the entire structural content of the drift. The intra-triple variation at scale P *is* the finest-scale content at that level of the label hierarchy: sibling triples differ exactly in the top trit of the branch target's label. The down-flow of fine modes is therefore starved by a factor $\kappa \le 1$ (with $\kappa < 1$ whenever the triples are not comonotone in their increments), while the up-flow is free. In the recurrence (7) this is exactly the statement $c > a$: the drift of the Collatz cascade is the statement that the minimum is a low-pass filter acting in one direction only. What remains is quantitative: how much attenuation, and is it uniform?

Definition 5 (Attenuation constant)**.** For sibling triples at scale P of the depth-k certificate, let D_i ($i = 1, 2, 3$) be the member difference-increments under one application of the edge-linearized operator and $D_{\min}$ the increment of the triple minimum. Set

$$\kappa(P; k) \;:=\; \frac{\mathrm{std}(D_{\min})}{\mathrm{std}(D_{\mathrm{member}})}.$$

Measured at $k = 13$ (certificate `cert_k13`): $\kappa(P) = 0.908,\ 0.899,\ 0.888,\ 0.875,\ 0.860,\ 0.841$ at $P = 2, \ldots, 7$ — decreasing with depth and converging into the range of the θ-series ($\to 0.849$). The identification $\theta_\infty = \lim \kappa$ holds to $\approx 1.2\%$; the residual gap is the up-channel mixing correction, to be accounted in the variational argument (Section 8).

5 Measured: the clipping decomposition and the mechanism

Where in the statistics does the attenuation live? Write

$$D_{\min} \;=\; \bar{D} + R, \qquad \bar{D} := \tfrac{1}{3}(D_1 + D_2 + D_3),$$

so R is the *min-correction*: the excess movement of the selected minimum over the triple average.

Proposition 6 (Clipping decomposition; verified, exact identity)**.** *With* $V := \mathrm{Var}(D_{\mathrm{member}})$*,*

$$\kappa^2 \;=\; \frac{\mathrm{Var}(\bar{D})}{V} \;+\; \frac{2\,\mathrm{cov}(\bar{D}, R)}{V} \;+\; \frac{\mathrm{Var}(R)}{V} \qquad \textit{exactly.} \tag{10}$$

Measured on `cert_k13`*,* $P = 2, \ldots, 7$*: the* entire *attenuation lives in the covariance term,* $2\mathrm{cov}/V = -0.170 \to -0.257$*, growing in magnitude with depth, while* $\mathrm{Var}(R)/V$ *is negligible (*0.010–0.028*).*

The correction R is *antisymmetric*:

$$\mathbb{E}[R \mid \bar{D} > 0] = -0.026, \qquad \mathbb{E}[R \mid \bar{D} < 0] = +0.026 \qquad (P = 7),$$

textbook one-sided clipping, as forced by the sandwich inequality

$$D_{\arg\min(\mathrm{new})} \;\le\; D_{\min} \;\le\; D_{\arg\min(\mathrm{old})}. \tag{11}$$

The effective law is linear mean reversion:

$$D_{\min} \;\approx\; (1-\lambda_{\mathrm{clip}})\, \bar{D} + \text{small noise}, \qquad \lambda_{\mathrm{clip}} := -\frac{\mathrm{cov}(\bar{D}, R)}{\mathrm{Var}(\bar{D})} = 0.085,\ 0.105,\ 0.122,\ 0.137 \tag{12}$$

at $P = 2, 4, 6, 7$ (measured; the full range over scales and resolutions is $\lambda_{\mathrm{clip}} = 0.085$–$0.140$, growing with depth). The attenuation constant is $\kappa \approx 1 - \lambda_{\mathrm{clip}}$ to leading order.

5.1 The mechanism is not switching

A natural guess is that the attenuation is produced by *switches* — events where the argmin changes member, clipping the increment via (11). The switch-resolved decomposition refutes this (measured, $P = 2, \ldots, 7$): the antisymmetric correction is *not* switch-exclusive. No-switch events carry the same ± 0.026 asymmetry as switch events, with $P(\text{switch}) \approx 0.66$ flat across scales. The mechanism is therefore *selection-weighted mean reversion*: the argmin member systematically co-moves less than the triple average, whether or not the selection changes hands.

5.2 LP interpretation: binding-constraint rigidity

The K–L system at the critical λ_k is a linear program, and the Perron eigenvector its solution; the min-selected member of each triple is the *binding* constraint of the row that reads it. Complementary slackness pins binding constraints into the active equation network, while slack members float freely with the perturbation field. The measured selection-weighted mean reversion is exactly what binding-constraint rigidity predicts: the bound member's response to a coarse perturbation is tied down by the equations it saturates, so it moves *less* than its free siblings — by the measured factor $1 - \lambda_{\mathrm{clip}}$.

The drift of the Collatz cascade is the rigidity of binding constraints in the Krasikov–Lagarias linear program.

This identification also dictates the proof route for the last link: LP duality. Show that the dual weights concentrate on binding entries, making their sensitivity to coarse perturbations strictly smaller than the free members' (Section 8).

6 Measured: uniformity of the attenuation across depth

The drift chain fails if $\kappa \to 1$ as $k \to \infty$ (triples becoming asymptotically comonotone). It does not:

Proposition 7 (Kappa uniformity; measured, four certificates)**.** *Across the certificates* $k = 13, 15, 17, 19$*:* $\kappa(P; k) < 0.95$ *at every scale* P *and every depth* k*; and the top-aligned deep-scale value is flat in* k*:*

$$\kappa_{\mathrm{deep}} = 0.841\ (k{=}13), \quad 0.840\ (15), \quad 0.837\ (17), \quad 0.838\ (19),$$

giving

$$\kappa_\infty \ = \ 0.839 \pm 0.002,$$

uniformly bounded away from 1*.*

Two remarks. First, the flatness in k is the empirically decisive point: every previously measured quantity of the program (temper exponent, θ, q) drifts with depth, but the attenuation does not — it is a *constant* of the cascade, not a running coupling. Second, the comparison with the profile root: θ-series $\to 0.849$ versus $\kappa_\infty = 0.839$; the identification $\theta = \kappa$ holds to $\approx 1.2\%$, the residual being the up-channel mixing correction. The last empirical link of the drift chain is pinned: the attenuation does not degenerate as depth grows.

The reason it cannot degenerate is itself already measured at nine depths: attenuation requires non-degenerate intra-triple variation, and the fine-end injection saturates at a *nonzero* limit.

Proposition 8 (Fine-end saturation; measured, nine depths [5])**.** *The finest-level triple* CV *of the certificate (each depth at its own critical* λ_k*) satisfies*

$$\mathrm{CV}_1(k) \ = \ 0.5136 \ - \ 0.337\,(0.910)^k$$

with residual 10^{-6} *across nine depths in* $8 \le k \le 20$*: geometric convergence to the finite, strictly positive limit* ≈ 0.514*.*

Intra-triple variation never vanishes; the raw material on which the low-pass acts is permanently supplied. (The rate 0.910 recurs across the program; see Appendix A.)

7 Negative results, honestly reported

Three natural shortcuts fail. We record them because each failure constrains the form any eventual proof can take.

7.1 The Gaussian reduction fails

One would like to compute κ from second-moment data alone: model the triple increments as (correlated) Gaussians with the measured covariance structure and evaluate the min-attenuation in closed form. This fails decisively (verified by Monte Carlo): Gaussian triples matched to the measured moments give $\kappa \approx 0.98$–1.00, versus the measured 0.84–0.91 — even when the measured leader–increment mean-reversion $\rho_{LI} = -0.085$ to -0.184 is built in. The attenuation constant is a *higher-order* statistic of the joint law of (levels, increments, selection); no second-moment surrogate reproduces it. Lesson: only the exact-identity route — working with the true certificate ensemble — gives the right constant.

7.2 One-step coefficient reconstruction inverts

One would like to derive the recurrence coefficients (a, c) of (7) directly as the bare per-application transfer weights of the linearized operator (the weighted channel masses of Section 3 adjusted by κ). This reconstruction produces coefficients with the *wrong ordering*: the one-step bare weights, read off naively, invert the drift direction. The effective (a, c) of the profile are renormalized, emergent quantities — as already signaled by the incoherence factor 0.90 in Corollary 2 and by the renormalized closure coefficients of the ultrametric spacing law [5]. The bare-to-effective map is nontrivial, and any proof that treats (7) as a literal one-step transfer law will prove the wrong thing.

7.3 Per-application mode transfer is scale-scattering

Strongest of the three: track a pure intra-triple mode at scale P through *one* application of the edge operator and measure the scale-decomposition of its output. The transfer is not tridiagonal — it is *scattering*: across all input scales P, fully 60–68% of any mode's output energy lands on digit 0 (the finest scale), regardless of P (measured). A single application does not trickle mass gently between neighboring scales; it slams most of it to the fine end, from which the stationary state rebuilds the profile.

The consequence is structural. The clean two-coefficient recurrence (7) — tridiagonal in scale, fit error $\leq 1.2\%$ — is *not* a per-application law. It is an *emergent, stationary* description: a property of the Perron fixed point, produced by the balance between per-application scattering and the stationary rebuild. Therefore the remaining proof cannot be iterative (following modes step by step); it must be *variational* — a statement about the fixed point itself. This is consistent with, and explains, the LP route of Section 5: the fixed point is an LP optimum, and LP optima are exactly the objects about which variational (duality, sensitivity) statements are natural.

8 The remaining lemma

Every structural link of the drift chain is now proved, and every quantitative link measured and uniform. Assembling Lemma 1 (proved), Proposition 3 (proved), Proposition 4 (proved), Proposition 7 (measured), and the edge rate $d\gamma/dq = 1/\ln(4/3)$ (proved), the single missing statement is:

Open Lemma 9 (Uniform attenuation at the fixed point)**.** *Let c_k be the Perron eigenvector of the critical K–L system $L_k(\lambda_k)$ and let $\kappa(P;k)$ be the attenuation constant of Definition 5, evaluated in the stationary state. There exists $\kappa_{\max} < 1$, independent of P and k, such that*

$$\kappa(P;k) \;\le\; \kappa_{\max} \qquad \textit{for all } P \textit{ and all } k,$$

provided the intra-triple coefficient of variation is bounded below,

$$\inf_k \mathrm{CV}_1(k) \;>\; 0$$

— which is precisely the nonzero saturation of Proposition 8 (measured: limit ≈ 0.514, geometric convergence, residual 10^{-6} over nine depths).

Given Open Lemma 9 (together with the bookkeeping that converts one-sided low-passing into $c - a > 0$ in the emergent recurrence — the bare-to-effective map of Section 7), the chain closes: $\theta \le \bar{\theta} < 1$ uniformly, hence $\mathrm{CV}_{\mathrm{top}}(k) \to 0$ geometrically, hence $q_k \to 1$, hence by the edge rate $\gamma_k \to 1$:

$$\pi_1(x) \;\ge\; x^{1-\varepsilon} \qquad \text{for every } \varepsilon > 0.$$

8.1 The LP-duality route

The negative results of Section 7 rule out moment-matching and iterative-transfer proofs; the mechanism identification of Section 5 points to the one framework in which the statement is natural. Sketch:

1. At the critical λ_k, the certificate is the optimum of the K–L linear program; each triple's argmin member is the binding entry of the row reading it, and complementary slackness attaches a strictly positive dual weight to each binding row.

2. Show the dual weights *concentrate* on binding entries: the dual solution assigns to each binding member a pinning strength bounded below, uniformly over rows, by the mass-conservation structure (Lemma 1 normalizes total dual mass; the balance identity, Proposition 3, prevents the dual mass from draining to either end of the scale hierarchy).

3. Sensitivity analysis: a coarse perturbation of the eigenvector moves each slack member with slope 1 (Proposition 4, coarse channel), while a binding member's response is reduced by its pinning strength. Non-degenerate triple gaps — guaranteed by $\inf \mathrm{CV}_1 > 0$ — keep the binding set stable under small perturbations, so the reduction is uniform: this is $\lambda_{\mathrm{clip}} \ge \lambda_{\min} > 0$, i.e. $\kappa \le 1 - \lambda_{\min} =: \kappa_{\max} < 1$.

Steps 1 and 3 are standard LP facts once step 2 is available; step 2 is where the specific arithmetic of the system ($2^\beta = 3$, the mass and balance identities) must enter. The measured target the argument must reproduce: $\kappa_\infty = 0.839 \pm 0.002$, λ_{clip} growing $0.085 \to 0.140$ with depth, and the ± 0.026 antisymmetric correction carried equally by switch and no-switch events.

9 Honest status

Proved	Mass conservation $3W_0 + W_2 + W_8 = 3$ and subcriticality $a + c \leq 1$ (Lem. 1, Cor. 2); balance identity, up-flow $= \frac{3}{16}$ = down-flow, zeroth-order drift zero (Prop. 3); directional low-pass $\min(x + c\mathbf{1}) = \min(x) + c$, 1-Lipschitz on fine modes, down-channel only (Prop. 4); edge rate $d\gamma/dq = 1/\ln(4/3)$ [5]; feasibility $\Rightarrow$ density bound [1].
Verified	Clipping decomposition (10) as an exact identity on `cert_k13`; sandwich inequality (11); pointwise transport identity (residual 2.8×10^{-4}) [5]; Gaussian Monte Carlo mismatch (Sect. 7).
Measured	$\kappa(P) = 0.908 \to 0.841$ at $k = 13$; covariance term $-0.170 \to -0.257$, $\mathrm{Var}(R)/V \leq 0.028$; antisymmetric correction ± 0.026, not switch-exclusive, $P(\text{switch}) \approx 0.66$; $\lambda_{\text{clip}} = 0.085$–$0.140$; $\kappa < 0.95$ everywhere, $\kappa_{\text{deep}} = 0.841/0.840/0.837/0.838$ at $k = 13/15/17/19$, $\kappa_\infty = 0.839 \pm 0.002$; $\theta = \kappa$ to 1.2%; fine-end saturation $\mathrm{CV}_1 \to 0.514 > 0$ (Prop. 8); scale-scattering 60–68% to digit 0; one-step reconstruction inversion; closure (7), fit error $\leq$ 1.2%; $a + c = 0.9955$ at $k = 20$.
Open	Open Lemma 9: $\kappa \leq \kappa_{\max} < 1$ uniformly at the fixed point, given $\inf \mathrm{CV}_1 > 0$; plus the bare-to-effective bookkeeping converting one-sided low-passing into $c - a > 0$.

The shape of the situation is unusual and worth stating plainly. The question "why do Collatz orbits fall?" has, at density level, been localized to a single mechanism with a single constant: the K–L certificate's heterogeneity cascade is pinned to the conservative line by the defining identity $2^\beta = 3$ (proved); its zeroth-order scale flow is exactly balanced by the same identity (proved); the only asymmetry is that the downward channel passes through a minimum, which transmits coarse structure perfectly and attenuates fine structure (proved); and the attenuation is measured to be a depth-independent constant, $\kappa_\infty = 0.839 \pm 0.002$, mechanistically identified as the rigidity of binding constraints in a linear program. Nothing in the chain is mysterious any longer; one variational inequality remains to be written down with full rigor, and the negative results show it must be proved at the fixed point, not by iteration.

A The constants of the program

The quantitative skeleton of the $\gamma \to 1$ program appears to be governed by a single constant: the drift of the map. Define

$$\delta := \log_2 \tfrac{16}{9} = 4 - 2\beta = 2\log_2 \tfrac{4}{3} = 0.830075\ldots, \tag{13}$$

twice the per-T-step $\log_2$ drift of the Collatz map (each odd step multiplies by 3/4 on average in $\log_2$: $\beta - 2 = -\delta/2$). Then (measured, except (iii) which is proved):

quantity	value	identification
$(a - c)$ renormalization flow rate	0.830	δ (rel. err. 0.01%)
fine-end saturation / homogenization rate	0.910	$\sqrt{\delta}$ (rel. err. 0.12%)
edge rate $d\gamma/dq$	3.47606...	$1/\ln(4/3)$ (closed form; proved)

The edge rate is obtained by implicit differentiation of the min-loss identity

$$1 = \lambda^{-2} + \tfrac{q}{3}\left(\lambda^{\beta-2} + \lambda^{\beta-1}\right)$$

at the edge $(\lambda, q) = (2, 1)$, giving exactly $d\gamma/dq = 1/\ln(4/3) = 3.47606\ldots$; the measured transfer ratios $(1-\gamma_k)/\big(3.4761\,(1-q_k)\big) = 0.824, 0.847, 0.873, 0.908$ at $k = 13, 15, 17, 19$ converge to 1, confirming first-order exactness [5]. The rate $\sqrt{\delta} = 0.9111$ appears in three measured roles — fine-end saturation (Prop. 8), per-depth homogenization, and the decay of $1 - q_k$ — one mechanism, three faces.

Two constants of the drift chain still lack convincing closed forms (honest gaps): $\theta_\infty \approx 0.8490$ (the candidate $27/32 = 0.84375$ misses by 0.6%, too coarse) and the saturation level $\mathrm{CV}_1^\infty \approx 0.5136$. The attenuation constant $\kappa_\infty = 0.839 \pm 0.002$ is numerically consistent with θ_∞ up to the measured 1.2% up-channel correction; whether κ_∞ itself is a function of δ is open.

Falsifiable predictions at the next depth ($k = 21$): flow rate $= \delta$ and saturation rate $= \sqrt{\delta}$ exactly (within measurement error); $\theta \approx 0.850$; $\kappa_{\mathrm{deep}} = 0.839 \pm 0.002$.

Reproducibility

All certificates, verification logs, and the scripts generating every number in this paper are archived in the research dossier accompanying this work (directories `certificates/`, `scripts/`, `findings/`; the measurements of Sections 5 and 6 are from the campaign records R766–R835).

Acknowledgment of method

The computations, fits, and drafts were produced in an extended human–AI collaboration; every proved statement above has a self-contained proof in the text, independent of that provenance.

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